DECODEMY

Foundations of Data Analytics · Pilot Program

Week 1:
Python Fundamentals for Data

Instructor: **Amos Augo** · Academy: **Decodemy**

Week 1 of 4 · 2 June 2026

By Friday, you will write real Python code that reads and analyzes data.

Week 1 Learning Objectives

By Friday, you will be able to...

- ✓ Create variables and work with Python's 4 core data types
- ✓ Use if/elif/else to make decisions in your code
- ✓ Build and manipulate lists to store collections of data
- ✓ Define reusable functions with parameters and return values
- ✓ Write for loops and while loops to automate repetitive tasks
- ✓ Read a real CSV file and calculate totals with Python

DECODEMY
LEARN. BUILD. SCALE.

Why Python for Data?

Task	Python	📊 Excel
1 million rows	Fast ⚡	Slow / crashes
Repeat weekly	1 script, 2 seconds	Manual clicking
Machine learning	5 lines of code ✅	Impossible
Cost	Free 🎉	$70/month
Scale	Unlimited ✅	~1M rows max

💡 **Key Point: "Python scales, Excel doesn't." Netflix, NASA, and Spotify all run on Python.**

The Data Analytics Workflow

Where Python fits in the bigger picture

Get Data	Clean	Analyze	Visualize	Present
CSVs, APIs, DBs	Fix errors, types	Stats & patterns	Charts & graphs	Reports & dashboards
Week 1 Python Basics	Week 1 Cleaning	Week 2 Pandas	Week 3 Tableau	Week 4 Capstone

📍 You are here — Week 1 builds the Python foundation for ALL of this.

DECODEMY
LEARN. BUILD. SCALE.

What is a Variable?

The building block of all data work

A Labeled Box

name = "Alex"
stores text "Alex"

A Price Tag

price = 19.99
stores a number

A Toggle

is_active = True
stores True/False

```
# Creating variables
name = "Alex"
sales = 250
tax_rate = 0.08
is_valid = True

print(f"Hello, {name}!")  # Hello, Alex!
```

⚠ Rules

- Use lowercase_with_underscores
- No spaces in names
- Case-sensitive: Sales ≠ sales
- Can't start with a number

Data Types - The 4 You Need

Every value in Python has a type

int

Whole numbers

```
age = 25
quantity = 3
```

float

Decimal numbers

```
price = 19.99
profit = -5.50
```

str

Text (in quotes)

```
name = "Alex"
region = 'North'
```

bool

True or False

```
is_active = True
on_sale = False
```

Your First Python Code 🎉

Run this in Jupyter — press Shift + Enter

```
print("Hello, Data! 👋")

# Try your own name
name = "Amos"
city = "Nairobi"
print(f"{name} is learning Python in {city}!")

# Output:
# Hello, Data! 👋
# Amos is learning Python in Nairobi!
```

⌨ **Shift+Enter**
Run a cell

+ **B**
New cell below

💾 **Ctrl+S**
Save notebook

■ **Esc+M**
Markdown cell

Lists – Storing Multiple Values

One variable, many items. Python counts from 0.

Positive:	0	1	2	3	4
Index:	100	250	75	400	120
Negative:	-5	-4	-3	-2	-1

```
prices = [100, 250, 75, 400, 120]

print(prices[0])    # 100  (first)
print(prices[1])    # 250  (second)
print(prices[-1])  # 120  (last)

prices.append(300)  # Add to end
print(len(prices))  # 6
```

Quick Reference

prices[0]	First item
prices[-1]	Last item
prices[1:3]	Slice (items 1 & 2)
len(prices)	Number of items
prices.append(x)	Add item
prices.remove(x)	Remove item

DECODEMY
LEARN. BUILD. SCALE.

Loops - Doing Things Repeatedly

What if you have 1,000 prices to process?

😩 Without Loops

```
print(prices[0])
print(prices[1])
print(prices[2]) ... forever
```

→

✅ With a For Loop

```
for price in prices:
    print(price)
```

```
# For loop – iterate through a list
prices = [100, 250, 75, 400, 120]

for price in prices:
    print(price)

# range() loop – repeat N times
for i in range(5):
    print(f"Row {i}")
```

While Loop

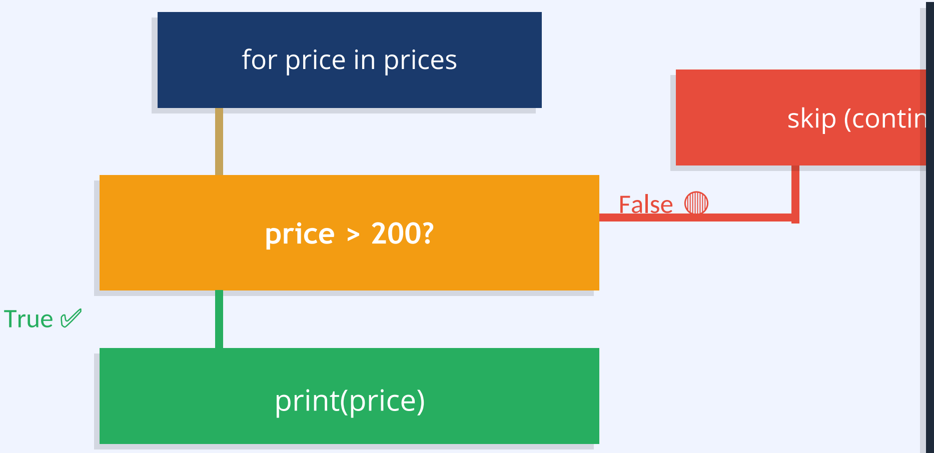
Use when you don't know
how many iterations you need.

```
count = 5
while count > 0:
    print(count)
    count -= 1
print("Blast off! 🚀")
```

DECODEMY
LEARN. BUILD. SCALE.

Conditionals – Making Decisions

Only process sales over $200




```
prices = [50, 250, 30, 500, 120, 1000]

for price in prices:
    if price > 200:
        print(f"High: ${price}")
    elif price > 100:
        print(f"Medium: ${price}")
    else:
        print(f"Low: ${price}")

# Operators: == != > < >= <=
```

Combining Loops + Conditionals

Real pattern: filter high-value orders

```
order_amounts = [50, 250, 30, 500, 120, 1000]

print("High-value orders (>$200):")

for amount in order_amounts:
    if amount > 200:
        print(f"  ✅ ${amount}")
```

Output:

```
High-value orders (>$200):
  ✅ $250    ✅ $500    ✅ $1000
```

🤔 Think-Pair-Share

What would happen if you changed > to < ?

Discuss with a neighbor → then type your answer in chat.

✏️ Pattern to remember

```
for item in collection:
    if condition:
        do_something()
```

DECODEMY LEARN. BUILD. SCALE.

Functions - Reusable Code

Define once, call many times

Input

price = 100
rate = 0.20

→

⚙ Function

apply_discount()

Does the calculation
every time you call it

→

Output

80.0

(20% off 100)

```
# DEFINE the function
def apply_discount(price, discount_rate):
    discounted = price * (1 - discount_rate)
    return discounted

# CALL it many times
print(apply_discount(100, 0.20))  # 80.0
print(apply_discount(250, 0.10))  # 225.0
print(apply_discount(400, 0.15))  # 340.0
```

Reading CSV Files - Real Data! 📂

CSV = Comma-Separated Values: the universal data format

📊 Spreadsheet view

product	qty	price
Laptop	2	899.99
Mouse	5	24.99
Keyboard	3	49.99

→

=

📄 Raw CSV text

```
product,qty,price
Laptop,2,899.99
Mouse,5,24.99
Keyboard,3,49.99
```

```python
import csv

with open('sales.csv', 'r') as file:
    reader = csv.reader(file)
    next(reader)       # Skip header row

    for row in reader:
        product  = row[0]
        quantity = int(row[1])    # string → int
        price    = float(row[2])  # string → float
        revenue  = quantity * price
```

DECODEMY
LEARN. BUILD. SCALE.

Week 1 Exercises 🚀

Two Jupyter notebooks to complete by Sunday

01	**Variables, Lists & Loops** Build a mini weekly sales tracker. Create a list of 7 daily sales, write a loop to print each, calculate total, average, best day.
02	**Functions + CSV Analysis** Write calculate_total_revenue() that reads sample_sales.csv and returns total revenue. You will read a REAL file and compute totals.

✔ **Push completed notebooks to GitHub by Sunday midnight**

Common Mistakes - Don't Panic ⚠

Every error is fixable. Read the message. It tells you exactly what's wrong.

IndentationError

"expected an indented block"

Fix: for price in prices:
    print(price) ← 4 spaces

💡 Python uses indentation instead of { } braces.

NameError

"name 'price' is not defined"

Fix: price = 100
print(price) ← create first!

💡 Create a variable before you use it.

TypeError

"can only concatenate str to str"

Fix: total = int("5") + 10 # = 15

💡 Convert strings to numbers before math.

IndexError

"list index out of range"

Fix: print(items[2]) # or items[-1]

💡 Python lists go from 0 to len-1.

Week 1 Schedule

Four touchpoints this week — show up and you will succeed

Tuesday	Thursday	Saturday	Sunday
Lecture	**Live Lab**	**Office Hours**	**Deadline**
2 Hours	1 Hour	1 Hour	Midnight
Python fundamentals, live demos, code-along. Slides + class notes provided.	Instructor-guided coding session. Work on Exercise 1 together in Jupyter.	11 AM ET. Drop-in session. Bring your stuck code. No question too small.	Submit both Jupyter notebooks via GitHub. Push by 11:59 PM.

DECODEMY
LEARN. BUILD. SCALE.

Success Looks Like...

By Sunday night, you will have all three of these:

✔ 2 Completed Notebooks

Exercise 1 (Variables/Loops) and Exercise 2 (Functions + CSV) saved and working.

✔ A Revenue Calculator

A Python function that opens a CSV file and computes total revenue — automatically.

✔ Code Pushed to GitHub

Your first professional code submission. Your portfolio starts today.

"You will feel like a real programmer. Because you will be one."

Preview: Week 2 - Pandas 🐼

What took 10 lines in Week 1 takes 1 line in Week 2

😄 Week 1 (Python)

```
import csv
total = 0
with open('sales.csv') as f:
    reader = csv.reader(f)
    next(reader)
    for row in reader:
        qty = int(row[1])
        price = float(row[2])
        total += qty * price
print(total)
```

→

🐼 Week 2 (Pandas)

```
import pandas as pd

df = pd.read_csv('sales.csv')
total = (df['qty'] *
         df['price']).sum()
print(total)

# Same result. 6 lines.
# And you can do SO much more.
```

🔥 You need to understand Week 1 to truly understand why Pandas is magical. Stay the course.

Q&A - Help Resources

You are never stuck alone. Here's where to get help.

Slack #python-help

Post your code + error message. Response within 24 hours.
Emoji your question 🙋

Office Hours

Every Saturday 11 AM – 12 PM ET. Drop-in, no appointment needed.

GitHub Issues

Submit code for review. Comment on classmates' work.
Build your profile.

Class Notes (this doc)

Saved as class_notes.md — cheat sheet at the back.
Reference anytime.

"No question is too small. Asking is the fastest path to understanding."

DECODEMY
LEARN. BUILD. SCALE.

You Belong Here.

"Week 1 is the hardest week.
By Friday, you'll have superpowers."

- Python is the world's most popular data language — you chose wisely
- Every professional you admire started exactly where you are today
- Confusion is part of learning. It means you're growing.
- Your cohort is your team — help each other, rise together